Ex. 2. Time Series

1.(a) Quadratic Trend Model

$$Y_t = b_0 + b_1 t + b_2 t^2 + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

Data:

Y	t	t^2
22	1	1
18	2	4
21	3	9
19	4	16
17	5	25
16	6	36
18	7	49
21	8	64
20	9	81
23	10	100
22	11	121
25	12	144

$$X = \begin{pmatrix} b_0 & b_1 & b_2 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ 1 & 5 & 25 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 8 & 64 \\ 1 & 9 & 81 \\ 1 & 10 & 100 \\ 1 & 11 & 121 \\ 1 & 12 & 144 \end{pmatrix}$$

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ 1 & 4 & 9 & 16 & 25 & 36 & 49 & 64 & 81 & 100 & 121 & 144 \end{pmatrix} \begin{pmatrix} & & \\ & & \\ & & \\ & & \\ & & \\ & & \\ & & \\ & & \\ & & \\ & & \\ & & \\ & & \end{pmatrix}$$

$$= \begin{pmatrix} 12 & 78 & 650 \\ 78 & 650 & 6084 \\ 650 & 6084 & 60710 \end{pmatrix}$$

Determinant of $X^T X$

$$|X^T X| = \begin{pmatrix} \begin{vmatrix} 650 & 6084 \\ 6084 & 60710 \end{vmatrix} & -\begin{vmatrix} 78 & 6084 \\ 650 & 60710 \end{vmatrix} & \begin{vmatrix} 78 & 650 \\ 650 & 6084 \end{vmatrix} \\ -\begin{vmatrix} 78 & 650 \\ 6084 & 60710 \end{vmatrix} & \begin{vmatrix} 12 & 650 \\ 650 & 60710 \end{vmatrix} & -\begin{vmatrix} 12 & 78 \\ 650 & 60710 \end{vmatrix} \\ \begin{vmatrix} 78 & 650 \\ 650 & 6084 \end{vmatrix} & -\begin{vmatrix} 12 & 650 \\ 78 & 6084 \end{vmatrix} & \begin{vmatrix} 12 & 78 \\ 78 & 650 \end{vmatrix} \end{pmatrix}$$

CS 扫描全能王

$$|X^TX| = \begin{vmatrix} 244644 & -780780 & 52052 \\ -780780 & 306020 & -22308 \\ 52052 & -22308 & 1716 \end{vmatrix} = 2290288$$

$$b = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix} = (X^TX)^{-1} X^TY \quad \left[(X^TX)^{-1} = |X^TX|\right]$$

where $X^TY = \begin{pmatrix} \Sigma Y \\ \Sigma tY \\ \Sigma t^2 Y \end{pmatrix} = \begin{pmatrix} 242 \\ 1624 \\ 13996 \end{pmatrix}$

$$\therefore \quad b_0 = \frac{244644(242) - 780780(1624) + 52052(13996)}{2290288}$$

$$= \frac{52572520}{2290288} = 22.954\dot{5} \simeq 22.95454$$

$$b_1 = \frac{-780780(242) + 306020(1624) - 22308(13996)}{2290288}$$

$$= \frac{-4195048}{2290288} = -1.831668 \simeq -1.8317$$

$$b_2 = \frac{52052(242) - 22308(1624) + 1716(13996)}{2290288}$$

$$= \frac{385528}{2290288} = 0.168332 \simeq 0.1683$$

$$\therefore \quad \hat{y} = 22.955 - 1.832t + 0.168t^2$$

CS 扫描全能王

1(b) Given Durbin Watson = 2.3913

From table $n=12$ $k'=2$ ($\because$ t and t^2)

$d_L = 0.812$ $\quad d_U = 1.579$

$4-d_L = 3.188$ $\quad 4-d_U = 2.421$

For $3.188 < d = 2.3913 < 2.421$,

$\therefore$ the test is inconclusive, hence do not reject H_0.

Thus, $\rho = 0$, there is no autocorrelation in the residuals.

1(c) Given t-test for $b_1 = -3.2301 < -3.169$

and t-test for $b_2 = 3.9642 > 3.169$

d.f. $= 12-2 = 10$ with $\alpha = 0.01$

critical value $|t_{0.005;10}| = 3.169$

$\therefore$ Reject H_0 : $\begin{cases} b_1 = 0 \\ b_2 = 0 \end{cases}$ and conclude that $\begin{cases} b_1 \neq 0 \\ b_2 \neq 0 \end{cases}$

Hence, both b_1 and b_2 are significant.

But $R^2 = 0.721164$, saying that t and t^2 cannot explain too much variation of the model. (It explains only 72.12 %.)

$$R^2_{adj} = 1 - \frac{(1-R^2)(n-1)}{n-k-1} = 1 - \frac{(1-0.721164)(12-1)}{12-2-1}$$

$$= 1 - 0.34079956 = 0.65920044 \simeq 0.6592$$

k is the no. of independent variables (i.e., t and t^2).

CS 扫描全能王

1.(d)

Y_t	$\hat{Y}_t$	e_t
22	21.2912	0.7088
18	19.9645	-1.9645
21	18.9746	2.0254
19	18.3212	0.6788
17	18.004	-1.004
16	18.024	-2.024
18	18.381	-0.381
21	19.0744	1.9256
20	20.1044	-0.1044
23	21.471	1.529
22	23.1743	-1.1743
25	25.9143	-0.9143

$$U = \frac{\sqrt{\frac{1}{n}\sum (Y_t - \hat{Y}_t)^2}}{\sqrt{\frac{1}{n}\sum \hat{Y}_t^2} + \sqrt{\frac{1}{n}\sum Y_t^2}} = \frac{\sqrt{\sum e_t^2}}{\sqrt{\sum \hat{Y}_t^2} + \sqrt{\sum Y_t^2}} = \frac{\sqrt{22.44602599}}{\sqrt{4972.09063} + \sqrt{4958}}$$

$$= \frac{4.7377237}{70.51305 + 70.413067} = \frac{4.7377237}{140.92611988} = 0.033618$$

which is quite close to 0, hence $\hat{Y}_t$ is a good forecasting model.

1(e) The model is quite acceptable judging from parts (c) and (d).

1(f) $\hat{y}_{13} = 22.9545 - 1.831668(13) + 0.168332(13)^2$

$= 27.590969$

$\simeq 27.591$

CS 扫描全能王

Table A-2

Models with an intercept (from Savin and White)

Durbin-Watson Statistic: 5 Per Cent Significance Points of dL and dU

	k'=1		k'=2		k'=3		k'=4		k'=5		k'=6		k'=7		k'=8		k'=9		
n	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL
6	0.610	1.400	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----
7	0.700	1.356	0.467	1.896	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----
8	0.763	1.332	0.559	1.777	0.367	2.287	----	----	----	----	----	----	----	----	----	----	----	----	----
9	0.824	1.320	0.629	1.699	0.455	2.128	0.296	2.588	----	----	----	----	----	----	----	----	----	----	----
10	0.879	1.320	0.697	1.641	0.525	2.016	0.376	2.414	0.243	2.822	----	----	----	----	----	----	----	----	----
11	0.927	1.324	0.758	1.604	0.595	1.928	0.444	2.283	0.315	2.645	0.203	3.004	----	----	----	----	----	----	----
12	0.971	1.331	0.812	1.579	0.658	1.864	0.512	2.177	0.380	2.506	0.268	2.832	0.171	3.149	----	----	----	----	----
13	1.010	1.340	0.861	1.562	0.715	1.816	0.574	2.094	0.444	2.390	0.328	2.692	0.230	2.985	0.147	3.266	----	----	----
14	1.045	1.350	0.905	1.551	0.767	1.779	0.632	2.030	0.505	2.296	0.389	2.572	0.286	2.848	0.200	3.111	0.127	3.360	----
15	1.077	1.361	0.946	1.543	0.814	1.750	0.685	1.977	0.562	2.220	0.447	2.471	0.343	2.727	0.251	2.979	0.175	3.216	0.
16	1.106	1.371	0.982	1.539	0.857	1.728	0.734	1.935	0.615	2.157	0.502	2.388	0.398	2.624	0.304	2.860	0.222	3.090	0.
17	1.133	1.381	1.015	1.536	0.897	1.710	0.779	1.900	0.664	2.104	0.554	2.318	0.451	2.537	0.356	2.757	0.272	2.975	0.
18	1.158	1.391	1.046	1.535	0.933	1.696	0.820	1.872	0.710	2.060	0.603	2.258	0.502	2.461	0.407	2.668	0.321	2.873	0.
19	1.180	1.401	1.074	1.536	0.967	1.685	0.859	1.848	0.752	2.023	0.649	2.206	0.549	2.396	0.456	2.589	0.369	2.783	0.
20	1.201	1.411	1.100	1.537	0.998	1.676	0.894	1.828	0.792	1.991	0.691	2.162	0.595	2.339	0.502	2.521	0.416	2.704	0.
21	1.221	1.420	1.125	1.538	1.026	1.669	0.927	1.812	0.829	1.964	0.731	2.124	0.637	2.290	0.546	2.461	0.461	2.633	0.
22	1.239	1.429	1.147	1.541	1.053	1.664	0.958	1.797	0.863	1.940	0.769	2.090	0.677	2.246	0.588	2.407	0.504	2.571	0.
23	1.257	1.437	1.168	1.543	1.078	1.660	0.986	1.785	0.895	1.920	0.804	2.061	0.715	2.208	0.628	2.360	0.545	2.514	0.
24	1.273	1.446	1.188	1.546	1.101	1.656	1.013	1.775	0.925	1.902	0.837	2.035	0.750	2.174	0.666	2.318	0.584	2.464	0.
25	1.288	1.454	1.206	1.550	1.123	1.654	1.038	1.767	0.953	1.886	0.868	2.013	0.784	2.144	0.702	2.280	0.621	2.419	0.
26	1.302	1.461	1.224	1.553	1.143	1.652	1.062	1.759	0.979	1.873	0.897	1.992	0.816	2.117	0.735	2.246	0.657	2.379	0.
27	1.316	1.469	1.240	1.556	1.162	1.651	1.084	1.753	1.004	1.861	0.925	1.974	0.845	2.093	0.767	2.216	0.691	2.342	0.
28	1.328	1.476	1.255	1.560	1.181	1.650	1.104	1.747	1.028	1.850	0.951	1.959	0.874	2.071	0.798	2.188	0.723	2.309	0.
29	1.341	1.483	1.270	1.563	1.198	1.650	1.124	1.743	1.050	1.841	0.975	1.948	0.900	2.052	0.826	2.164	0.753	2.278	0.
30	1.352	1.489	1.284	1.567	1.214	1.650	1.143	1.739	1.071	1.833	0.998	1.931	0.926	2.034	0.854	2.141	0.782	2.251	0.
31	1.363	1.496	1.297	1.570	1.229	1.650	1.160	1.735	1.090	1.825	1.020	1.920	0.950	2.018	0.879	2.120	0.810	2.226	0.
32	1.373	1.502	1.309	1.574	1.244	1.650	1.177	1.732	1.109	1.819	1.041	1.909	0.972	2.004	0.904	2.102	0.836	2.203	0.
33	1.383	1.508	1.321	1.577	1.258	1.651	1.193	1.730	1.127	1.813	1.061	1.900	0.994	1.991	0.927	2.085	0.861	2.181	0.
34	1.393	1.514	1.333	1.580	1.271	1.652	1.208	1.728	1.144	1.808	1.079	1.891	1.015	1.978	0.950	2.069	0.885	2.162	0.
35	1.402	1.519	1.343	1.584	1.283	1.653	1.222	1.726	1.160	1.803	1.097	1.884	1.034	1.967	0.971	2.054	0.908	2.144	0.
36	1.411	1.525	1.354	1.587	1.295	1.654	1.236	1.724	1.175	1.799	1.114	1.876	1.053	1.957	0.991	2.041	0.930	2.127	0.
37	1.419	1.530	1.364	1.590	1.307	1.655	1.249	1.723	1.190	1.795	1.131	1.870	1.071	1.948	1.011	2.029	0.951	2.112	0.
38	1.427	1.535	1.373	1.594	1.318	1.656	1.261	1.722	1.204	1.792	1.146	1.864	1.088	1.939	1.029	2.017	0.970	2.098	0.
39	1.435	1.540	1.382	1.597	1.329	1.658	1.273	1.722	1.218	1.789	1.161	1.859	1.104	1.932	1.047	2.007	0.990	2.085	0.
40	1.442	1.544	1.391	1.600	1.338	1.659	1.283	1.721	1.230	1.786	1.175	1.854	1.120	1.924	1.064	1.997	1.008	2.072	0.
45	1.475	1.566	1.430	1.615	1.383	1.666	1.336	1.720	1.287	1.776	1.238	1.835	1.189	1.895	1.139	1.958	1.089	2.022	1.
50	1.503	1.585	1.462	1.628	1.421	1.674	1.378	1.721	1.335	1.771	1.291	1.822	1.246	1.875	1.201	1.930	1.156	1.986	1.
55	1.529	1.601	1.490	1.641	1.452	1.681	1.414	1.724	1.374	1.768	1.334	1.816	1.294	1.861	1.253	1.909	1.212	1.959	1.
60	1.549	1.616	1.514	1.652	1.480	1.689	1.444	1.727	1.408	1.767	1.372	1.809	1.335	1.850	1.298	1.894	1.260	1.939	1.
65	1.567	1.629	1.536	1.662	1.503	1.696	1.471	1.731	1.438	1.767	1.404	1.802	1.370	1.843	1.336	1.882	1.301	1.923	1.
70	1.583	1.641	1.554	1.672	1.525	1.703	1.494	1.735	1.464	1.768	1.433	1.802	1.401	1.838	1.369	1.876	1.337	1.916	1.
75	1.598	1.652	1.571	1.680	1.543	1.709	1.515	1.739	1.487	1.770	1.458	1.801	1.429	1.834	1.399	1.867	1.369	1.901	1.
80	1.611	1.662	1.586	1.688	1.560	1.715	1.534	1.743	1.507	1.772	1.480	1.801	1.453	1.831	1.425	1.861	1.397	1.893	1.
85	1.624	1.671	1.600	1.696	1.575	1.721	1.550	1.747	1.525	1.774	1.500	1.801	1.474	1.829	1.448	1.857	1.422	1.886	1.
90	1.635	1.679	1.612	1.703	1.589	1.726	1.566	1.751	1.542	1.776	1.518	1.801	1.494	1.827	1.469	1.854	1.445	1.881	1.
95	1.645	1.687	1.623	1.709	1.602	1.732	1.579	1.755	1.557	1.778	1.535	1.803	1.512	1.827	1.489	1.852	1.465	1.877	1.
100	1.654	1.694	1.634	1.715	1.613	1.736	1.592	1.758	1.571	1.780	1.550	1.803	1.528	1.826	1.506	1.850	1.484	1.874	1.
150	1.720	1.747	1.706	1.760	1.693	1.774	1.679	1.788	1.665	1.802	1.651	1.817	1.637	1.832	1.622	1.848	1.608	1.862	1.
200	1.758	1.779	1.749	1.789	1.739	1.799	1.729	1.809	1.718	1.820	1.707	1.831	1.697	1.841	1.686	1.852	1.675	1.863	1.

*'k' is the number of regressors excluding the intercept